

Maqsudur Rahman

Ph.D. Candidate in Computer Science, Boise State University, USA Expected Ph.D. Completion: **May 2027**

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Professional Profile

Computer Science educator and Ph.D. Candidate with **7+ years of university-level teaching experience**, prior appointments as **Assistant Professor, Lecturer, and Senior Lecturer**, and active research in **trustworthy AI, secure machine learning, quantum machine learning, medical/biomedical AI, and hybrid quantum-classical systems**. Experienced in Bachelor-level teaching, laboratory instruction, student advising, curriculum delivery, undergraduate project/thesis supervision, and research mentoring. Research and teaching profile aligns with KFUPM areas including **information security, secure computing, software engineering, high-performance computing, bioinformatics, and information visualization**.

Education

Boise State University, Boise, Idaho, USA Aug 2024–Present
Ph.D. Candidate in Computer Science; Coursework CGPA: **3.89/4.00**; Advisor: Dr. Jun Zhuang.
Research Areas: Trustworthy AI, Quantum Neural Networks, Robust Medical Imaging, Secure/Adversarial AI.
Comilla University, Bangladesh 2015–2017
M.Sc. (Engineering) in Information and Communication Technology; CGPA: **3.96/4.00** (Rank: 1st).
B.Sc. (Engineering) in Information and Communication Technology; CGPA: **3.87/4.00** (Rank: 1st).

Academic Appointments

Assistant Professor, Department of ICT, Comilla University, Bangladesh Nov 2023–Jul 2024
Lecturer, Department of ICT, Comilla University, Bangladesh Nov 2021–Oct 2023
Senior Lecturer, Department of CSE, Port City International University, Bangladesh Jan 2020–Oct 2021
Lecturer, Department of CSE, Port City International University, Bangladesh Apr 2017–Dec 2019
Research Assistant, Department of ICT, Comilla University, Bangladesh Jul 2014–Jun 2016

Teaching, Mentoring, and Student Supervision

- Taught a broad range of undergraduate Computer Science and ICT courses: **Structured Programming Language, Object-Oriented Programming Language, Data Structures, Algorithms, Operating Systems, Information Security, Artificial Intelligence, Machine Learning, and Deep Learning**.
- Supervised **56 undergraduate projects** and **15 thesis students**: 40 projects and 8 thesis students at Port City International University; 16 projects and 7 thesis students at Comilla University.
- Supported student-centered learning through laboratory instruction, project-based learning, academic advising, research mentoring, and programming activities.
- Coached the CoU “Unpredictable 3207” ACM ICPC team, which achieved back-to-back divisional champion recognition.

Research and Publication Highlights

- Publication record includes **26 peer-reviewed or accepted scholarly works**, including Ph.D.-stage work in **MICCAI, AAAI, and Journal of Imaging**, plus one **NeurIPS 2026 manuscript currently under review**.
- Ph.D. research develops robust quantum-enhanced and trustworthy AI models for medical image classification under **noisy labels, corrupted inputs, weak supervision, backdoor poisoning, and quantum trainability challenges**.
- **NQNN (MICCAI 2025)**: noise-aware quantum neural network for multiclass medical image classification under 10%–50% symmetric label noise.
- **QAPNet (AAAI 2026)**: quantum-attentive patchwise network for robust medical image classification under noisy imaging inputs.
- **Journal of Imaging 2026**: published work on barren-plateau mitigation in QNNs and systems-level support for hybrid quantum-classical medical imaging pipelines.
- Earlier research includes machine learning, NLP, cyberbullying detection, intrusion detection, VANET security, spam detection, and intelligent data analysis.

Research and Teaching Interests Aligned with KFUPM

Artificial Intelligence and Machine Learning; Information Security and Secure Computing; Secure/Trustworthy Machine Learning; Software Engineering and AI Systems; High-Performance and Quantum Computing for AI; Bioinformatics, Clinical AI, and Medical Image Analysis; Information Visualization for Biomedical Data; Data Security and Adversarial Robustness.

Grants, Honors, and Academic Service

- **Principal Investigator**, UGC-funded project on Bangla cyberbullying detection using PMI and machine learning, Comilla University, 2022–2023.
- **Principal Investigator**, UGC-funded project on Bangla cyberbullying corpus creation and detection using adaptive neuro-fuzzy inference systems, Comilla University, 2021–2022.
- Best Paper Award Finalist, ICAIAA 2022; Best Paper Award Finalist, IEEE ICAECT 2021.
- M.Sc. highest CGPA recognition; B.Sc. distinction and university merit scholarship.
- Reviewer for peer-reviewed journals and international conferences; moderator/advisory roles in science, IT, and ICT student organizations.

Technical Skills

Programming: Python, C, C++, Java. **AI/ML:** PyTorch, TensorFlow, Scikit-learn, Keras, Hugging Face Transformers.

Quantum Computing: PennyLane, Qiskit, variational quantum circuits, quantum neural networks, cuQuantum.

Trustworthy/Secure AI: noisy-label learning, uncertainty estimation, adversarial robustness, backdoor poisoning defense.

Data/Medical Imaging: OpenCV, NumPy, Pandas, SciPy, Matplotlib, MedMNIST, biomedical image classification.

Tools: L^AT_EX, Git/GitHub, Jupyter, Microsoft Office, Overleaf.